

Cohort Analysis: User Retention Rate

https://docs.google.com/spreadsheets/d/1tN_7WJ2VTtBQ-3DI753lt3VQIXsOxbpXhTEGVuLSACY/edit?usp=sharing

Project Overview

This project focuses on evaluating user retention rates using cohort analysis methodology through SQL and Google Sheets. The analysis compares the behavior of promotional users versus organic users to assess acquisition quality and retention patterns over a 6-month observation period.

Objectives

- Apply cohort analysis principles to calculate user retention rates
- Extract and transform data from a database using SQL
- Build interactive cohort tables in Google Sheets
- Compare retention patterns between promotional and organic user segments
- Formulate data-driven conclusions about acquisition quality

Tech Stack

- Database: PostgreSQL
- SQL Client: DBeaver (or alternative)
- Visualization: Google Sheets (Pivot Tables, Conditional Formatting, Slicers)
- Query Language: SQL

Dataset Structure

Table 1: cohort_users_raw

Contains baseline user information:

Field	Description
user_id	Unique user identifier
full_name	User's full name
email	User email (may contain noise)
country	User's country
signup_datetime	Registration date/time (mixed text formats)
signup_source	Acquisition channel

signup_device Device type

promo_signup_flag 1 = promotional user, 0 = organic user

Note: **signup_datetime** requires cleaning due to inconsistent formats (varying delimiters: ., /, - and date component lengths).

Table 2: cohort_events_raw

Tracks user activity post-registration:

Field	Description
event_id	Unique event identifier
user_id	Associated user
event_datetime	Event timestamp (mixed formats)
event_type	Event type (login, view_content, purchase, registration, etc.)
revenue	Transaction amount (if applicable)

Data Characteristics:

- First event per user is always registration
- Inconsistent date formats requiring normalization
- May contain test events, NULL values, or out-of-window events

Project Tasks

Task 1: Database Work & Cohort Table Preparation

Tools: DBeaver / SQL Client

Key Steps:

- 1. Date Normalization**
 - Clean and standardize **signup_datetime** and **event_datetime** fields
 - Handle multiple date formats (delimiters, 2-digit vs 4-digit years)
 - Convert text dates to proper timestamp format using **CASE** statements
- 2. Data Transformation**
 - Create CTEs for cleaned user and event data
 - Join tables on **user_id**

- Calculate cohort month (YYYY-MM format from registration date)
- Calculate month_offset (user tenure in months)
- 3. Data Filtering
 - Remove records with NULL dates
 - Exclude NULL event_type values
 - Filter out test events (event_type = 'test_event')
 - Limit observation window to January-June 2025
- 4. Final Aggregation
 - Group by: promo_signup_flag, cohort_month, month_offset
 - Calculate: unique user counts per cohort/month combination
 - Sort by promo flag, cohort month, then month offset

Output Format:

promo_signup_flag	cohort_month	month_offset	users_total
0	2025-01-01	0	400
0	2025-01-01	1	303
0	2025-01-01	2	210

Task 2: Building Cohort Tables in Google Sheets

Tools: Google Sheets

Key Steps:

1. Data Import
 - Import SQL results into Google Sheets
 - Verify data format settings
2. Cohort Table Construction
 - Create pivot table with:
 - Rows: cohort_month
 - Columns: month_offset
 - Values: users_total (unique count)
 - Apply gradient conditional formatting
3. Retention Rate Calculation
 - Build parallel table with same structure
 - Calculate retention rate relative to month 0 (cohort baseline)
 - Format as percentages (month 0 = 100%)
 - Ensure empty cells align with base table
4. Interactive Elements
 - Create slicer for promo_signup_flag filtering
 - Verify dynamic updates across both tables
5. Analysis & Conclusions
 - Write comparative analysis of promo vs. organic cohorts
 - Document retention patterns and acquisition quality insights

Key Deliverables

- 1. SQL Script: Working query producing the aggregated cohort table**
- 2. Google Sheets Workbook with 3 sheets:**
 - **Data: Imported SQL results**
 - **Cohort_tables: Interactive pivot tables with slicer**
 - **Conclusions: Written analysis**

Skills Demonstrated

- **SQL data cleaning and transformation**
- **Date parsing with mixed formats**
- **CTE and JOIN operations**
- **Cohort analysis methodology**
- **Retention rate calculation**
- **Google Sheets pivot tables and conditional formatting**
- **Data visualization and storytelling**
- **Comparative behavioral analysis**

Analysis Focus

This project evaluates:

- **Promotional Users: Acquired through campaigns/advertising**
- **Organic Users: Natural/direct acquisition**
- **Retention Patterns: How each segment retains over 6 months**
- **Acquisition Quality: Which channel produces more engaged long-term users**